IT2011
AI and ML

Lecture 02 (Part 1)

Expert Systems and Ontologies

Year 02
Semester 01

"Learn Today
Apply Tomorrow !" ☺

1 What is an Expert System ?

- An Expert System is a computer program that simulates the decision-making ability of a human expert in a specific domain.
- It uses expert knowledge, facts and rules to give advice, solve problems or make decisions.
- It is designed to behave like a human expert.
- Expert systems are part of Artificial Intelligence (AI).

" An Expert System captures the knowledge and experience of a human expert and uses it to solve real-world problems. " ☺

2 Key Characteristics

- ✓ Uses domain-specific knowledge
- ✓ Based on facts and IF-THEN rules
- ✓ Provides advice, explanations and solutions
- ✓ Can handle uncertain or incomplete information
- ✓ Works like a human expert
- ✓ Useful in specialized areas
- ✓ Consistent and reliable decisions
- ✓ Can explain how it reached a conclusion

Tries to think like a human expert ! ☺

" Expert systems do not replace human experts, but they support and assist them. " ☺

3 Examples of Expert Systems

Domain	Example	Task
Healthcare	MYCIN	Diagnosis of bacterial infections
Agriculture	CROP-EXPERT	Suggest crops and fertilizers
Automotive	Auto-Expert	Troubleshoot car problems
Finance	Financial Expert	Investment advice
Education	Intelligent Tutor	Provide personalized tutoring
Industry	Fault Diagnosis	Find faults in machines
General	ChatGPT (Generative AI)	Answer questions, give advice

" Expert systems are used in many fields where human expertise is valuable ! " ☺

4 Data → Information → Knowledge

Data (Raw Facts)
- Raw facts and figures.
- No meaning by itself.
- Examples: numbers, text, images, etc.

32
37
38
...

→

Information (Processed Data)
- Data that is organized and meaningful.
- Helps to understand a situation.
- Example: "The temperature today is 37°C."

Temperature today is 37°C

→

Knowledge (Understanding)
- Gained from information and experience.
- Used for decision making.
- Example: "If temperature is high, there may be a fever."

High temperature may indicate a fever

" Data becomes useful information, and information leads to knowledge. " ☺

5 Expert vs Novice

Human Expert	Expert System
• Has years of experience	• Uses stored knowledge and rules
• Uses knowledge and intuition	• Gives consistent answers
• Can handle uncertain situations	• Can work 24/7
• May be inconsistent	• Does not have real understanding
• May not always be available	• Limited to its knowledge base
• Can explain reasoning (usually)	• Explains reasoning (if designed to)

6 Advantages of Expert Systems

- ✓ Preserve expert knowledge
- ✓ Provide consistent and reliable decisions
- ✓ Available anytime (24/7)
- ✓ Can be used in dangerous or remote environments
- ✓ Can help less experienced people
- ✓ Useful in many real-world applications
- ✓ Can explain the reasoning behind decisions
- ✓ Reduce human error

Same quality advice every time ! ☺

7 Limitations (Briefly)

- ✗ Limited to the knowledge it has
- ✗ Cannot think or learn like a human (unless combined with ML)
- ✗ Difficult to capture expert knowledge
- ✗ May not work well in completely new situations
- ✗ Can be expensive and time-consuming to develop

Expert systems are powerful, but they also have limitations ! ☺

Key Takeaway !

1. Expert systems simulate human expertise.
2. They use knowledge, facts and rules.
3. Data → Information → Knowledge.
4. They are useful but have limitations.

" Knowledge makes better decisions possible ! " ☺

IT2011
AI and ML

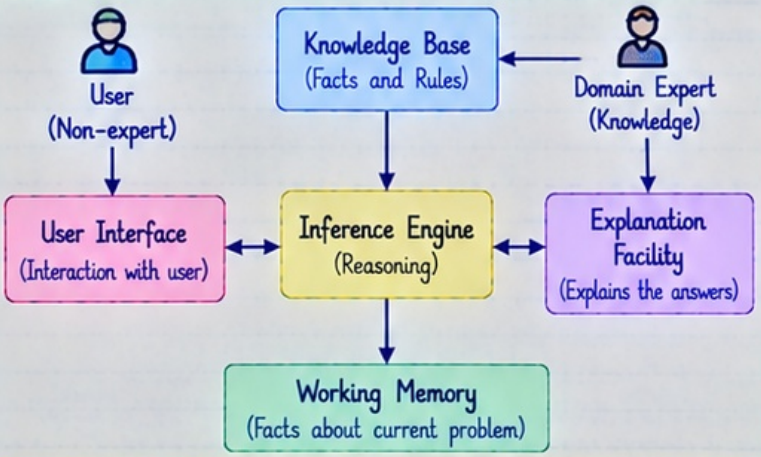
Lecture 02 (Part 2)

Expert System Architecture and Components

Year 02
Semester 01

"Understand the parts
Build the big picture !" ☺

1 Main Components of an Expert System



All components work together to solve a problem like a human expert !" ☺

2 Description of Components

Component	Function (Simple Explanation)
Knowledge Base	• Stores domain knowledge (facts and rules). • Usually in the form of IF-THEN rules.
Inference Engine	• Uses rules and facts to derive new facts and reach a conclusion. • Can use forward chaining or backward chaining.
Working Memory	• Stores facts about the current problem. • Updated during the reasoning process.
User Interface	• Allows the user to enter information. • Displays the results in a simple way.
Explanation Facility	• Explains how the system reached the answer. • Increases trust and understanding.
Domain Expert	• A human expert who provides the knowledge. • Helps to build and maintain the knowledge base.

3 Knowledge Representation

- Knowledge in expert systems is represented using:
  - Facts (simple statements)
  - Rules (IF-THEN rules)
- The most common form is the production rule:
  IF condition(s) THEN conclusion.
- Example:
  IF fever is high AND cough is present
  THEN the patient may have flu.
- Rules are provided by a domain expert.
- The knowledge base can be updated when new knowledge is available.
- Good knowledge representation is important for accurate reasoning.

Knowledge is stored in a way that the computer can understand ! ☺

4 Example of Facts and Rules

Facts (Examples)

- Patient has fever.
- Patient has cough.
- Temperature is high.
- Car engine is not starting.
- Soil is dry.
- Leaves are yellow.

Rules (IF-THEN)

- IF fever is high AND cough is present THEN illness is flu.
- IF engine is not starting AND battery is low THEN the problem is a dead battery.
- IF soil is dry AND leaves are yellow. THEN the plant needs water.
- IF customer is a student THEN give a discount.

Facts describe the current situation, and rules tell the system what to do with those facts. ☺

5 Simple Example: Medical Diagnosis

Knowledge Base (Rules)

R1: IF fever is high AND cough is present THEN illness is flu.
R2: IF headache is present AND fever is high THEN illness is dengue.
R3: IF sore throat is present THEN illness is cold.

Facts (Given by User)

- Fever is high.
- Cough is present.

Inference Engine

Conclusion

Illness is flu.

Explanation Facility

"Fever is high and cough is present, so rule R1 is used to conclude flu."

The expert system gives a decision and also explains how it reached that decision !" ☺

6 Key Points to Remember

1. Expert systems use domain-specific knowledge.
2. Knowledge is represented using facts and IF-THEN rules.
3. Main components: knowledge base, inference engine, working memory, user interface, explanation facility.
4. Forward chaining starts from facts and moves to a conclusion.
5. Backward chaining starts from a goal and works backwards.
6. Explanation facility increases trust and helps users understand.
7. Expert systems are useful in many real-world applications.
8. They are not as flexible as human experts.
9. The quality of the system depends on the knowledge provided.
10. Ontologies help to organize and represent knowledge in a structured way (discussed later).

Revise Regularly ! ☺

" Understand the components well, it will help you in the exam !" ☺

" Knowledge + Reasoning + Explanation = A Useful Expert System !" ☺

Keep Going !

IT2011
AI and ML

Lecture 02 (Part 3)

Inference Engine and Reasoning in Expert Systems

Year 02
Semester 01

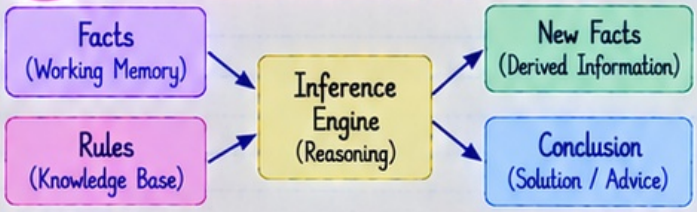
"Think like an expert, Reason step by step !"

1 What is an Inference Engine ?

- The inference engine is the brain of the expert system.
- It uses rules from the knowledge base and facts from working memory to derive new facts and reach a conclusion.
- It decides which rule to apply, in what order, and how to use the facts.
- It can use forward chaining or backward chaining.
- It continues the reasoning process until a solution is found or no more rules can be applied.

" Inference engine applies knowledge to facts to get new facts and reach a conclusion. "

2 How the Inference Engine Works ?



Steps:

1. Check the available facts.
2. Find a rule that matches the facts.
3. Apply the rule and derive a new fact.
4. Add the new fact to working memory.
5. Repeat the process until a conclusion is reached.
6. If no rule can be applied, stop.

It keeps thinking step by step !

3 Forward Chaining (Data-Driven)

- Starts with known facts and moves forward to a conclusion.
- Checks which rules can be applied using the current facts.
- When a rule is applied, a new fact is added.
- The process continues until the goal (conclusion) is reached or no more rules can be applied.
- Useful when we have data and want to find what can be concluded.

Example:

R1: IF fever AND cough THEN flu
R2: IF flu THEN rest
R3: IF flu THEN see a doctor

Working:

Facts: fever, cough
↓
Apply R1 → flu
↓
Apply R2 → rest
↓
Apply R3 → see a doctor

Starts with facts and moves forward !

Forward chaining is like solving a puzzle step by step from the facts we know. "

4 Backward Chaining (Goal-Driven)

- Starts with a goal (what we want to prove) and works backwards to find the required facts.
- Checks which rules can give the goal.
- Then it tries to prove the conditions (sub-goals) of those rules.
- This process continues until all facts are found or the goal cannot be proven.
- Useful when we have a specific goal and want to check if it is true.

Example:

Goal: flu

R1: IF fever AND cough THEN flu

Working:

Goal: flu
↓
Check R1 (need fever AND cough)
↓
Are fever and cough known?
↓
Yes → Goal proven (flu)

Starts with a goal and works backwards !

Backward chaining is like asking "What do I need to know to prove this? "

5 Comparison: Forward vs Backward Chaining

Aspect	Forward Chaining	Backward Chaining
Starting point	Known facts (data-driven)	Goal (hypothesis-driven)
Direction	Forward to conclusion	Backward to facts
When to use	When we have data and want a conclusion	When we have a specific goal to check
Search strategy	Breadth-first search	Depth-first search
May generate irrelevant facts	Yes	No (focuses on goal)
Example use	Monitoring, prediction	Diagnosis, verification

Both methods are useful. The choice depends on the problem and situation.

6 Other Reasoning Methods (Briefly)

- Forward and backward chaining are the most common methods, but there are other reasoning techniques too.
- Examples: fuzzy reasoning, probabilistic reasoning, case-based reasoning.
- These methods are useful for handling uncertainty, incomplete information and real-world problems.

" Different reasoning methods help expert systems work in more complex and uncertain situations. "

7 Key Takeaways

- Inference engine is the heart of the expert system.
- Forward chaining starts with facts.
- Backward chaining starts with a goal.
- Both methods use rules to derive new facts.
- The reasoning method depends on the problem.
- Understanding these concepts is important for exams and real-world use.

Understand → Practise → Apply → Succeed !

Keep Learning Keep Growing !

" Reasoning turns facts into solutions ! "

IT2011
AI and ML

Lecture 02 (Part 4)

Ontologies in Artificial Intelligence

Year 02
Semester 01

"Organize knowledge for a smarter world !"

1 What is an Ontology ?

- An ontology is a structured way of representing knowledge in a domain.
- It defines the important concepts (classes), their properties (attributes) and the relationships between them.
- It helps computers (and humans) to have a common understanding of the meaning of terms.
- Ontologies are widely used in AI, semantic web, expert systems and knowledge-based Systems.

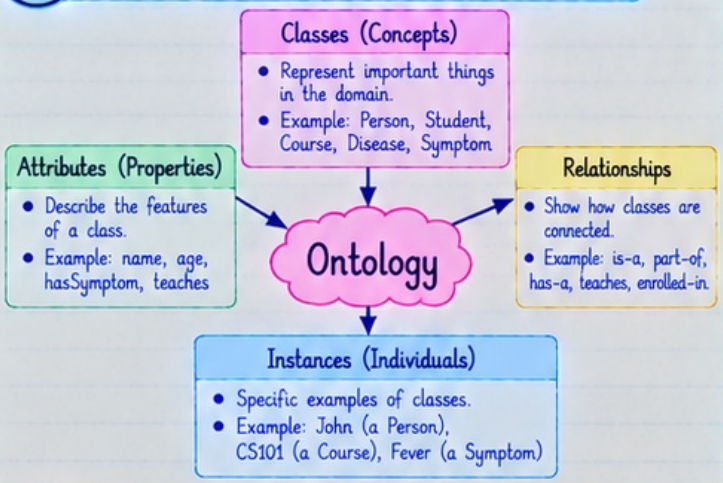
" An ontology is like a dictionary for a specific domain, but more powerful because it also shows how the words (concepts) are connected ! "

2 Why are Ontologies Important ?

- Help to share and reuse knowledge.
- Provide a common understanding of terms.
- Reduce ambiguity (same word can have different meanings).
- Useful in communication between humans and machines.
- Make it easier to organize, search and reason about knowledge.
- Support interoperability between different systems.
- Used in many real-world applications (e.g. healthcare, education, e-commerce, etc.).

" Ontologies help different people and systems to understand the same thing in the same way. "

3 Main Components of an Ontology

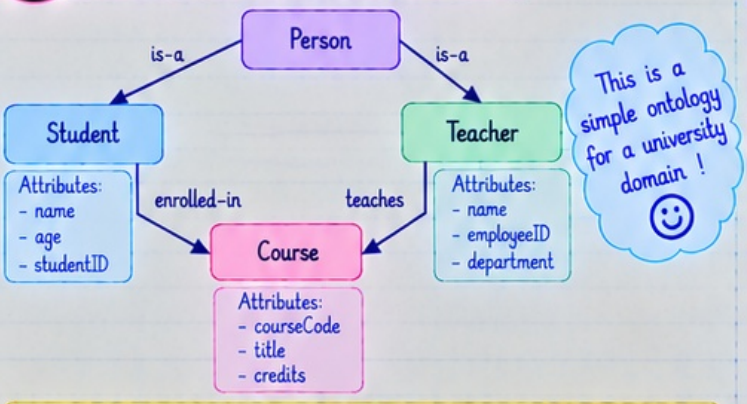


4 Types of Relationships

Relationship Type	Meaning	Example
is-a (subclass)	A class is a type of another class	Student is-a Person
has-a (part-of)	A class has a part or attribute	Car has-a Engine
part-of	A class is part of another class	Engine part-of Car
related-to	General relationship	Student related-to Course
instance-of	An individual belongs to a class	John instance-of Student
equivalent-to	Two classes mean the same thing	Automobile equivalent-to Car
disjoint-with	Two classes cannot be the same	Student disjoint-with Teacher

" Relationships connect the concepts and give meaning to the ontology. "

5 Example: University Ontology



This ontology shows the main concepts, their attributes and relationships in a simple university domain.

6 Real-World Applications

Domain	How Ontologies are Used	Example
Healthcare	Represent diseases, symptoms, treatments	Disease ontology (e.g. SNOMED CT)
Education	Represent courses, skills, learning resources	E-learning ontology
E-commerce	Represent products, categories, attributes	Product ontology
Transportation	Represent vehicles, routes, locations	Smart transport systems
Semantic Web	Organize web knowledge for better search	Search engines (e.g. Google)
Expert Systems	Structure domain knowledge for reasoning	Medical diagnosis systems

" Ontologies are used wherever knowledge needs to be organized and shared effectively. "

7 Key Takeaways

- Ontology = structured representation of knowledge.
- Main components: classes, attributes, relationships, instances.
- Different types of relationships (is-a, has-a, part-of, etc.).
- Ontologies reduce ambiguity and help sharing knowledge.
- Used in many real-world applications.
- Important for AI, expert systems and the semantic web.

Learn
Understand
Apply
Succeed !

Remember !

1. Classes = main concepts
2. Attributes = describe concepts
3. Relationships = connect concepts
4. Instances = real examples
5. Ontologies = common understanding

Keep Going !

" A well-structured ontology helps build smarter AI systems ! "

IT2011
AI and ML

Lecture 02 (Part 5)

Ontology Development and Real-World Use

Year 02
Semester 01

"Knowledge organized today, smarter AI tomorrow!"

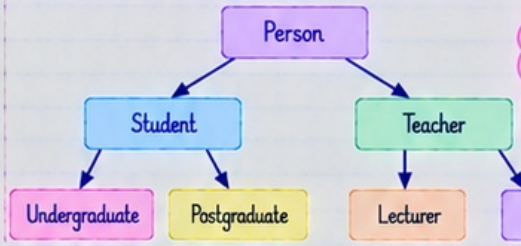
1 Steps to Build an Ontology

1. **Define the domain and scope**
   - Decide what area the ontology will cover (e.g. university, medical).
2. **Collect important terms (concepts)**
   - Find the main concepts, attributes and relationships.
3. **Organize the concepts into classes**
   - Arrange the concepts in a hierarchy (general → specific).
4. **Define the attributes (properties)**
   - Describe the features of each class.
5. **Define the relationships**
   - Specify how the classes are connected (e.g. is-a, has-a).
6. **Create instances (individuals)**
   - Add real examples of the classes.
7. **Validate and refine**
   - Check for mistakes and improve the ontology.

Build step by step and keep it simple!

2 Example: University Ontology (Detailed)

Class Hierarchy (is-a relationship)



Classes are organized from general to specific!

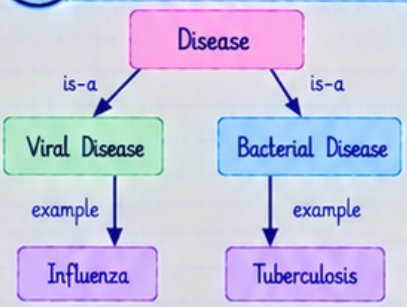
Attributes (Properties)

- Person: name, age, email
- Student: studentID, course, year
- Teacher: employeeID, department
- Course: courseCode, title, credits

Relationships

- Student is-a Person
- Teacher is-a Person
- Student enrolled-in Course
- Teacher teaches Course
- Course has-a Department

3 Another Example: Medical Ontology



Attributes (Example for Disease)

- name
- symptoms
- treatment
- cause
- severity

Relationships (Examples)

- Disease has-symptom Symptom
- Disease has-treatment Treatment
- Symptom indicates Disease

"Ontologies can be used in healthcare to organize medical knowledge and help in diagnosis, research and treatment."

4 Tools for Building Ontologies

Tool	Description	Features	Used For
Protégé	Most popular ontology editor	Free, open source, user-friendly	Creating and editing ontologies
OWL	Web Ontology Language	Standard language for ontologies	Representing ontologies on web
RDF	Resource Description Framework	Used to represent data about resources	Storing and sharing ontology data
SPARQL	Query language for RDF	Used to search and retrieve data from ontologies	Querying ontology data

"Protégé is widely used because it is easy to use and powerful."

5 Advantages and Challenges of Ontologies

Advantages

- Provide a common understanding of terms.
- Help in organizing and sharing knowledge.
- Reduce ambiguity (same word, different meanings).
- Improve communication between humans and machines.
- Useful in many real-world applications (e.g. healthcare, education, e-commerce).
- Support interoperability between different systems.

Challenges

- Time-consuming to develop.
- Requires domain expertise.
- Can be complex for large domains.
- Keeping the ontology updated can be difficult.
- Different people may have different opinions on how to organize knowledge.
- May not cover all possible situations.

"Ontologies are powerful, but they require careful design and maintenance."

6 Real-World Applications of Ontologies

Domain	Example Use
Healthcare	Disease ontology (e.g. SNOMED CT) for diagnosis and research.
Education	E-learning ontology to organize courses and learning resources.
E-commerce	Product ontology to classify products and attributes.
Transportation	Ontology for routes, vehicles and locations.
Semantic Web	Organize web knowledge for better search and information retrieval.
Expert Systems	Domain ontology to support reasoning and decision making.

"Ontologies help make information more useful, connected and accessible in the real world."

7 Key Takeaways

1. Ontology = structured knowledge representation.
2. Main components: classes, attributes, relationships, instances.
3. Use is-a, has-a, part-of and other relationships.
4. Follow a clear process to build an ontology.
5. Ontologies are used in many real-world applications.
6. Tools like Protégé, OWL, RDF and SPARQL are commonly used.
7. They have many advantages but also some challenges.

Understand
Practice
Apply
Succeed!

For the Exam!

- Be able to define an ontology.
- Explain the components of an ontology.
- Give examples of classes, attributes and relationships.
- Understand different types of relationships.
- Know the steps to build an ontology.
- Explain real-world applications.
- Discuss advantages and challenges.
- Try to draw simple diagrams.

Keep Revising! You Can Do It!

"A well-designed ontology turns information into knowledge, and knowledge into a smarter future!"

IT2011
AI and ML

Lecture 02 (Part 6)

Advanced Topics in Ontologies and Summary

Year 02
Semester 01

"Build knowledge,
Build a smarter world !"

1 Ontology Languages in More Detail

- There are several languages used to create and represent ontologies.
- The most popular is OWL (Web Ontology Language).
- OWL is based on XML and has different versions:

OWL Versions:

- OWL Lite – simpler, for basic ontologies
- OWL DL – more expressive, supports reasoning
- OWL Full – most expressive, but less computable

- Other related languages: RDF and RDFS.
- These languages help to represent knowledge in a standard and machine-readable way.

"OWL is the most widely used language for building ontologies."

2 Ontology Development Process (Life Cycle)

1. Specification → Define purpose, scope and requirements.
2. Conceptualization → Identify key concepts, attributes and relationships.
3. Formalization → Represent the ontology using a language (e.g. OWL).
4. Implementation → Build the ontology using tools (e.g. Protégé).
5. Evaluation → Check if the ontology is correct and useful.
6. Maintenance → Update and improve over time.

"Building an ontology is an iterative process."

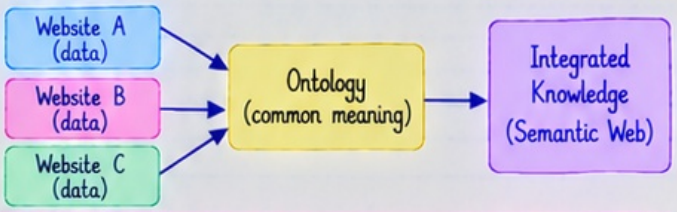
3 Ontology vs Database (Comparison)

Aspect	Ontology	Database
Purpose	Represent knowledge and meaning	Store data
Structure	Rich semantics (classes, properties, relationships)	Tables with fixed structure
Flexibility	High (can be extended easily)	Fixed schema (harder to change)
Understanding	Machine and human readable (with meaning)	Mainly for machines
Example	OWL, RDF	MySQL, SQL Server

"A database stores data, but an ontology gives meaning to data."

4 Use of Ontologies with the Semantic Web

- The Semantic Web is an extension of the current web.
- It uses ontologies to provide meaning to the information on the web.
- Ontologies help search engines and applications to better understand content.
- They enable data from different sources to be connected and integrated.
- Examples: Google Knowledge Graph, linked data, open data initiatives.



"Ontologies make the web smarter by giving meaning to information."

5 Current Trends and Future of Ontologies

- Ontologies are increasingly used in AI, big data, IoT and knowledge graphs.
- They help in automatic reasoning, data integration and intelligent search.
- Domain ontologies are being developed for:
  - Healthcare (e.g. SNOMED CT)
  - Education (e.g. learning ontologies)
  - Smart cities (e.g. sensor data)
  - E-commerce (e.g. product ontology)
  - Agriculture, environment, finance, etc.
- In the future, ontologies will play a key role in building more intelligent and connected systems.

Ontologies are the building blocks of a smarter future !

6 Final Summary and Key Takeaways

- An ontology is a structured representation of knowledge in a domain.
- It consists of classes, attributes, relationships and instances.
- Ontologies reduce ambiguity and improve communication.
- Common relationship types: is-a, has-a, part-of, etc.
- Popular tools: Protégé, OWL, RDF, SPARQL.
- Ontologies are widely used in healthcare, education, e-commerce, semantic web and expert systems.
- They follow a step-by-step development process.
- Ontologies help machines and humans understand information in the same way.
- They are an important part of modern AI and the Semantic Web.

"Ontologies turn information into knowledge and help build smarter AI systems."

"Knowledge Organized Today, Smarter Solutions Tomorrow !"

Keep Learning !

IT2011
AI and ML

Lecture 02 (Part 7)

Past Questions, Exam Focus and Revision

Year 02
Semester 01

"Revise Smartly, Score Confidently!" ☺

1 Common Exam Questions (Short Answers)

1. What is an expert system ?
2. What is an ontology ?
3. Differentiate between data, information and knowledge.
4. List the main components of an expert system.
5. What is an inference engine ?
6. Explain forward chaining with a simple example.
7. Explain backward chaining with a simple example.
8. What are the main parts of an ontology ?
9. What is an ontology language ? Give examples.
10. What are the advantages of using ontologies ?

"Practising past questions helps you understand what is really important!" ☺

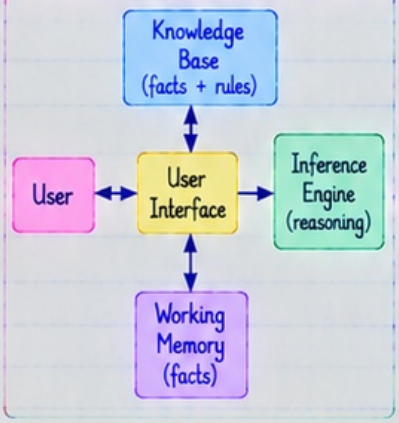
2 Possible Long Answer Questions

1. Explain the architecture of an expert system with a neat diagram and describe each component.
2. Discuss forward and backward chaining with examples. Compare the two methods.
3. Explain the concept of ontology in detail. How are ontologies used in artificial intelligence ?
4. Describe the ontology development process with a diagram.
5. Discuss the real-world applications of ontologies in different domains.

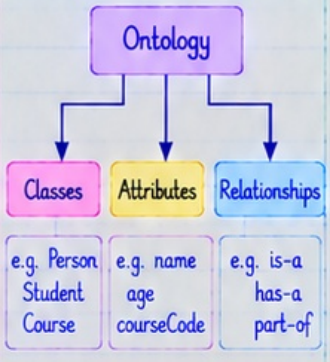
"For long answers, use diagrams, examples and clear explanations to get higher marks." ☺

3 Diagrams to Remember

Expert System Architecture



Ontology Structure



"Diagrams help you remember the structure and relationships clearly." ☺

4 Key Comparison Tables

Forward vs Backward Chaining

Aspect	Forward Chaining	Backward Chaining
Starting point	Known facts	Goal / hypothesis
Direction	Forward (data-driven)	Backward (goal-driven)
When to use	When we have data and want a conclusion	When we have a specific goal to check
Search strategy	Breadth-first	Depth-first
May generate irrelevant facts	Yes	No (focuses on goal)
Example use	Monitoring, prediction	Diagnosis, verification

Ontology vs Database

Aspect	Ontology	Database
Purpose	Represent meaning	Store data
Structure	Rich semantics	Fixed schema
Flexibility	High	Low
Understanding	Machine + human readable	Mainly for machines

5 Quick Revision Checklist

- ✓ Understand the definition of expert systems and ontologies.
- ✓ Know the difference between data, information and knowledge.
- ✓ Be able to draw and explain the expert system architecture.
- ✓ Understand forward and backward chaining with examples.
- ✓ Know the components of an ontology (classes, attributes, relationships, instances).
- ✓ Be familiar with ontology development steps.
- ✓ Know popular tools (Protégé, OWL, RDF, SPARQL).
- ✓ Understand real-world applications (healthcare, education, e-commerce, etc.).
- ✓ Practise past questions.
- ✓ Be able to explain advantages and challenges.

Check every point before the exam! ☺

6 Study Tips for This Lecture

- Read your notes regularly (small portions).
- Practise drawing diagrams from memory.
- Try to explain each concept in your own words.
- Do past papers and time yourself.
- Focus more on understanding, not just memorizing.
- Use examples to remember concepts.
- Discuss with friends and teach someone else.
- Keep a summary sheet for final revision.

Study Plan + Practice + Succeed! ☺

"A well-prepared mind always finds a better way!" ☺

All the best
You can do it! ☺